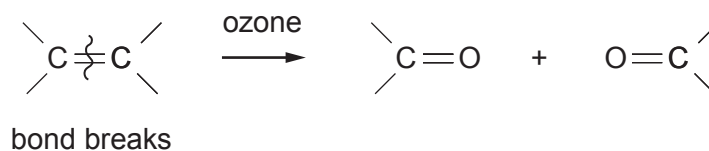


9. (a) Alkenes react with ozone to give an intermediate compound, which can then react further to give aldehydes or ketones.



1 mol of an alkene **T** reacts with ozone to give 2 mol of the same carbonyl compound, **U**.

- (i) Explain what can be deduced about the structural formula of alkene **T**. [1]

.....

.....

- (ii) Carbonyl compound **U** reacted with 2,4-dinitrophenylhydrazine to produce a solid derivative.

State the colour of this solid derivative. [1]

.....

- (iii) I. The ^1H NMR spectrum of compound **U** did **not** show a chemical shift at 9.8δ .

State what can be deduced from this statement. [1]

.....

.....



- II. The melting temperatures of the 2,4-dinitrophenylhydrazine derivatives can be used to identify the original carbonyl compound. The melting temperatures of some of these derivatives are shown in the table.

Carbonyl compound	Melting temperature of derivative / °C
$\text{CH}_3(\text{CH}_2)_3\text{C} \begin{array}{l} \text{CH}_3 \\ \diagup \\ \text{C} \\ \diagdown \\ \text{O} \end{array}$	106
$\text{CH}_3(\text{CH}_2)_3\text{C} \begin{array}{l} \text{H} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{O} \end{array}$	108
$\text{CH}_3\text{CH}_2\text{C} \begin{array}{l} \text{CH}_3 \\ \diagup \\ \text{C} \\ \diagdown \\ \text{O} \end{array}$	115
$\text{CH}_3(\text{CH}_2)_2\text{C} \begin{array}{l} \text{H} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{O} \end{array}$	126

The 2,4-dinitrophenylhydrazine derivative of compound **U** was slightly impure and melted at a temperature of 110-113 °C.

Use the information given to deduce the formula of compound **U**.
Explain your reasoning.

[3]

.....

.....

.....

.....

.....

.....

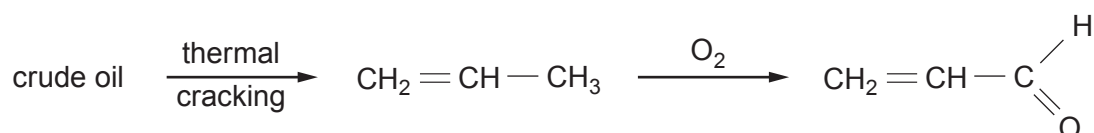
- (iv) Use your answer to part (iii) and other relevant information given in the question to give the structure of alkene **T**. Name alkene **T**. [2]

Name

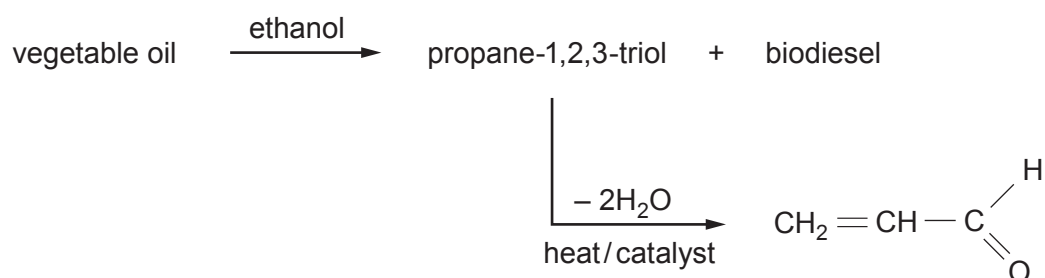


(b) Propenal, $\text{CH}_2 = \text{CH} - \text{CHO}$ is made by two main methods.

Method 1 From crude oil



Method 2 From vegetable oil



- (i) Suggest **one** reason why the general public would probably favour the production of propenal by Method 2. [1]

.....

.....

- (ii) If you were considering which of these two methods to choose to produce propenal, state **two** other factors, apart from cost, that you would need to know to be able to come to a decision. [2]

Factor 1

.....

Factor 2

.....



- 11
-
- I410U401

Liquid	Density / g cm ⁻³
biodiesel	0.85
propane-1,2,3-triol	1.26

Sketch and name a piece of apparatus that could be used to separate a mixture containing 25 cm³ of each liquid. The two liquids should also be labelled. [2]

Name of apparatus

- (iv) Propenal reacts with hydrogen cyanide by a nucleophilic addition reaction.

Complete the mechanism for this reaction using curly arrows and partial/complete charges as appropriate. [2]

